

LIE ALGEBRAS

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Upshot. Semisimple finite-dimensional Lie algebras can be classified using root space decomposition. We choose a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$ (see Definition 6.2) and consider the generalized weight decomposition with respect to the adjoint representation (which is by convention called the *root* space decomposition)

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_{\alpha} \quad \text{with } \mathfrak{g}_0 = \mathfrak{h},$$

where $[\mathfrak{g}_{\alpha}, \mathfrak{g}_{\beta}] \subset \mathfrak{g}_{\alpha+\beta}$.

“For arbitrary Lie algebras, Cartan subalgebras can have unwieldy structure, and an unpredictable action under the adjoint representation. We see from the previous ([Kac10, Lecture 12]) result, however, that with the assumption of semisimplicity the situation is much more transparent. We can rewrite the root space decomposition as

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_{\alpha}$$

where Δ is the set of all $\alpha \neq 0$ such that $\mathfrak{g}_{\alpha} \neq 0$. We call $\alpha \in \Delta$ a *root* of $\mathfrak{g}$, and $\mathfrak{g}_{\alpha}$ the corresponding *root space*.”

In Lecture 17 we have: “Previously, given a semisimple Lie algebra $\mathfrak{g}$ we constructed its associated root system (V, Δ) . (The construction depends on choosing a Cartan subalgebra, but by Chevalley’s theorem, the root systems constructed from the same $\mathfrak{g}$ are isomorphic.) Next, given a root system we’ll construct a *Cartan matrix* A , and from this we’ll eventually see how to reconstruct $\mathfrak{g}$.

We’ll see that to every root system corresponds a semisimple Lie algebra so it’s important to know all the root systems. Last time we saw the four series A_r, B_r, C_r and D_r , and the three exceptions E_6, E_7 and E_8 . The remaining two exceptions are F_4 and G_2 .”

Suppose (V, Δ) is a root system and $f : V \rightarrow \mathbb{R}$ is a linear map such that $f(\alpha) \neq 0$ for all $\alpha \in \Delta$. Define [Kac10, Definition 17.1]

- (1) $\alpha \in \Delta$ is *positive* if $f(\alpha) > 0$ and *negative* if $f(\alpha) < 0$.
- (2) A positive root is *simple* if it cannot be written as the sum of two positive roots.
- (3) A *highest root* $\theta \in \Delta$ is a root where f is maximal; that is $f(\theta) \geq f(\alpha)$ for all $\alpha \in \Delta$.

Finally, on Lecture 21, “Let V be a finite dimensional real vector space with a positive definite bilinear form $(\cdot, \cdot)$. Let $a \in V$ be a nonzero vector and denote by r_a the orthogonal reflection relative to a , i.e. $r_a(a) = -a$, $r_a(v) = v$ if $(a, v) = 0$, $r_a(v) = v - 2\frac{(a, v)}{(a, a)}a$.

Let (V, Δ) be a root system. Let W be the subgroup of $O(V)$ generated by $r_\alpha, \alpha \in \Delta$. The group W is called the *Weyl group* of (V, Δ) (and of the corresponding semisimple Lie algebra $\mathfrak{g}$.)”

1. BASIC DEFINITIONS AND EXAMPLES

Definition 1.1. An *algebra* is a vector space over a field $\mathbb{F}$ endowed with a binary operation that is bilinear

$$\begin{aligned} a(\lambda b + \mu c) &= \lambda ab + \mu ac \\ (\lambda b + \mu c)a &= \lambda ba + \mu ca \end{aligned}$$

Definition 1.2. A *Lie algebra* is an algebra $\mathfrak{g}$ with a product $[\cdot, \cdot]$ we call *bracket* that satisfies

- (1) $[x, x] = 0 \quad \forall x \in \mathfrak{g}$,
- (2) (Jacobi identity.)

Definition 1.3. A *simple Lie algebra* is a Lie algebra that has no nontrivial proper ideals.

As a rule of thumb I keep in mind the following silly computation:

$$\log \det A = \log \prod_i \lambda_i = \sum \log \lambda_i = \text{“trlog } A\text{”}$$

And recall that exponent map goes from $T_e G = \mathfrak{g} \rightarrow G$, so that logarithm would go from $G \rightarrow \mathfrak{g}$. This is why I remember that the condition on a classical Lie group of *having determinant 1* goes to *having vanishing trace* in the Lie algebra. (Because $\log 1 = 0$.)

Example 1.4. (1) The *special linear Lie algebra*

$$\mathfrak{sl}_n = \{\text{Mat}_n | \text{Tr}(A) = 0\}$$

which is just obvious from the slogan above.

(2) The *special orthogonal Lie algebra*

$$\mathfrak{so}_n = \{A \in \text{Mat}_n | A + A^T = 0\}$$

which is obvious from: $\text{SO}(n)$ = isometries, so $\langle v, v \rangle = \langle Av, Av \rangle = \langle v, A^T A v \rangle$, so $\text{SO}(n) = \{A \in \text{Mat}_n : A^{-1} = A^T\}$, and then $0 = \log 1 = \log(AA^T) = \log A + \log A^T$.

(3) The *symplectic Lie algebra*

$$\mathfrak{sp}_{2n} = \{A \in \text{Mat}_{2n} : \Omega A + A^T \Omega = 0\}$$

where $\Omega = \begin{pmatrix} 0 & \text{Id}_n \\ -\text{Id}_n & 0 \end{pmatrix}$.

Pending. Why this makes sense?

Example 1.5. If A is an associative Lie algebra, then A with the bracket $[a, b] = ab - ba$ is a Lie algebra, denoted by A_- . (It is an exercise to verify Jacobi's identity.) This gives $\mathfrak{gl}_V := \text{End}(V)_-$.

Definition 1.6. A *Lie subalgebra* is a vector subspace $\mathfrak{h} \subset \mathfrak{g}$ such that $[\mathfrak{h}, \mathfrak{h}] \subset \mathfrak{h}$.

Definition 1.7. A subspace $\mathfrak{h} \subset \mathfrak{g}$ is called an *ideal* if $[\mathfrak{h}, \mathfrak{g}] \subset \mathfrak{h}$.

Recall that a bilinear form $(\cdot, \cdot) : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ is *invariant* if $([a, b], c) = (a, [b, c])$ for all $a, b, c \in \mathfrak{g}$.

Definition 1.8. For any finite dimensional Lie algebra $\mathfrak{g}$ and $x \in \mathfrak{g}$ we write

$$\text{ad}_x = [x, -]$$

Exercise 1.9. Show that the map $\text{ad} : \mathfrak{g} \rightarrow \text{End}(\mathfrak{g})$, $a \mapsto \text{ad}_a$ is a representation, i.e. a morphism of Lie algebras.

Definition 1.10. The *Killing form* is

$$\kappa(x, y) = \text{Tr}_{\mathfrak{g}} \text{ad}_x \text{ad}_y$$

Exercise 1.11. Prove that an invariant bilinear form on a simple Lie algebra must in fact be symmetric.

Proof. (David.) It's enough to show that $\mathfrak{g}$ is *perfect*, i.e. that $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. In this case, let $a, b \in \mathfrak{g}$ and suppose that $b = [x, y]$. Then

$$\begin{aligned} (a, b) &= (a, [x, y]) = (a, -[y, x]) = (-[a, y], x) = ([y, a], x) \\ &= (y, [a, x]) = (y, -[x, a]) = (-[y, x], a) = ([x, y], a) = (b, a) \end{aligned}$$

To confirm that $\mathfrak{g}$ is perfect just observe that $[\mathfrak{g}, \mathfrak{g}]$ is a nontrivial ideal of $\mathfrak{g}$. $\square$

Definition 1.12. A *semisimple Lie algebra* is a direct sum of simple ones

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \dots \oplus \mathfrak{g}_s$$

Theorem 1.13 (Cartan). *The finite dimensional Lie algebra $\mathfrak{g}$ is semisimple if and only if its Killing form is nondegenerate.*

2. NILPOTENT AND SOLVABLE LIE ALGEBRAS

Let $\mathfrak{g}$ be a Lie algebra. Define

$$\mathfrak{g}^1 := [\mathfrak{g}, \mathfrak{g}], \quad \mathfrak{g}^2 := [\mathfrak{g}, \mathfrak{g}^1], \quad \dots, \quad \mathfrak{g}^n := [\mathfrak{g}, \mathfrak{g}^{n-1}]$$

and

$$\mathfrak{g}^{(1)} := [\mathfrak{g}, \mathfrak{g}], \quad \mathfrak{g}^{(2)} := [\mathfrak{g}^{(1)}, \mathfrak{g}^{(1)}], \dots, \quad \mathfrak{g}^{(n)} := [\mathfrak{g}^{(n-1)}, \mathfrak{g}^{(n-1)}].$$

Definition 2.1. A Lie algebra $\mathfrak{g}$ is

- *nilpotent* if there exists n such that $\mathfrak{g}^n = 0$,
- *solvable* if there exists n such that $\mathfrak{g}^{(n)} = 0$.

3. LIE'S THEOREM

Lie's theorem says that solvable Lie algebras over closed fields of characteristic not zero have weights.

Definition 3.1. Let $\mathfrak{h}$ be a Lie algebra, $\pi : \mathfrak{h} \rightarrow \mathfrak{gl}_V$ a representation of $\mathfrak{h}$ and $\lambda \in \mathfrak{h}^*$. The *weight space* of $\mathfrak{h}$ attached to λ is

$$V_\lambda^\mathfrak{h} := \{v \in V \mid \pi(h)v = \lambda(h)v, \forall h \in \mathfrak{h}\}.$$

If $V_\lambda^\mathfrak{h} \neq 0$, we say that λ is a *weight* for π .

Theorem 3.2 (Lie's theorem). *Let $\mathfrak{g}$ be a solvable Lie algebra and π a representation of $\mathfrak{g}$ on a finite dimensional vector space $V \neq 0$, over an algebraically closed field $\mathbb{F}$ of characteristic 0. Then there exists a weight $\lambda \in \mathfrak{g}^*$ for π (that is, $V_\lambda^\mathfrak{g} \neq \{0\}$).*

Exercise 3.3. Show the following two corollaries of Lie's theorem:

- for all representations π of a solvable Lie algebra $\mathfrak{g}$ on a finite dimensional vector space V over an algebraically closed field $\mathbb{F}$, $\text{char}\mathbb{F} = 0$, there exists a basis for V for which the matrices of $\pi(\mathfrak{g})$ are upper triangular;
- a solvable subalgebra $\mathfrak{g} \subset \mathfrak{gl}_V$, (where V is finite-dimensional over an algebraically closed field $\mathbb{F}$, $\text{char}\mathbb{F} = 0$), is contained in the subalgebra of upper triangular matrices over $\mathbb{F}$ for some basis of V .

4. GENERALIZED EIGENSPPACES AND GENERALIZED WEIGHT SPACES

Definition 4.1. A *generalized eigenspace* of $A \in \text{End}(V)$ with *eigenvalue* λ is

$$V_\lambda = \{v \in V : (A - \lambda \text{Id})^N = 0 \text{ for some positive integer } N\}.$$

It turns out that any linear operator on an algebraically closed field gives a decomposition into generalized eigenspaces via Jordan canonical form:

Proposition 4.2. *Let A be a linear operator on a finite-dimensional vector space V over an algebraically closed field $\mathbb{F}$, and let $\lambda_1, \dots, \lambda_s$ be all eigenvalues of A , and $n_1, \dots, n_s$ their multiplicities. Then one has the generalized eigenspace decomposition:*

$$V = \bigoplus_{i=1}^s V_{\lambda_i}, \quad \dim V_{\lambda_i} = n_i.$$

In particular, for a Lie algebra $\mathfrak{g}$ with a representation (i.e. Lie algebra morphism) $\pi : \mathfrak{g} \rightarrow \text{End}(V)$ we have

$$V = \bigoplus_{\lambda \in \mathbb{F}} V_\lambda^a, \quad V_\lambda^a = \{v \in V : (\pi(a)v - \lambda \text{Id})^N v = 0 \text{ for some } N \in \mathbb{N}\}$$

And even more particularly, for the adjoint representation we have

$$V = \bigoplus_{\lambda \in \mathbb{F}} \mathfrak{g}_\lambda^a, \quad \mathfrak{g}_\lambda^a = \{b \in \mathfrak{g} : ([a, \cdot] - \lambda \text{Id})^N b = 0, \text{ for some } N \in \mathbb{N}\}$$

And a little more generally we have

Definition 4.3. Let $\mathfrak{h}$ be a Lie algebra with a representation π on a vector space V , and $\lambda \in \mathfrak{h}^*$. A *generalized weight space* of $\mathfrak{h}$ in V attached to λ is

$$V_\lambda^\mathfrak{h} = \left\{ v \in V : (\pi(a) - \lambda(a)\text{Id})^N v = 0, \begin{array}{l} \text{for some } N \in \mathbb{N} \\ \text{depending on } a \in \mathfrak{h} \\ \text{for all } a \in \mathfrak{h} \end{array} \right\}$$

Under the right conditions, a nilpotent subalgebra $\mathfrak{h} \subset \mathfrak{g}$ permits decomposing V as a direct sum of generalized weight spaces of $\mathfrak{h}$. Namely,

Theorem 4.4. *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra and π its representation on a finite-dimensional vector space V , over an algebraically closed field $\mathbb{F}$ of characteristic 0. Let $\mathfrak{h}$ be a nilpotent subalgebra of $\mathfrak{g}$. Then the following equalities hold:*

$$(4.4.1) \quad \begin{aligned} V &= \bigoplus_{\lambda \in \mathfrak{h}^*} V_{\lambda}^{\mathfrak{h}} \\ \pi(\mathfrak{g}_{\alpha}^{\mathfrak{h}}) V_{\lambda}^{\mathfrak{h}} &\subseteq V_{\lambda+\alpha}^{\mathfrak{h}} \end{aligned}$$

Which in the case of the adjoint representation, looks like:

Definition 4.5. The *generalized root space decomposition* of a Lie algebra $\mathfrak{g}$ over an algebraically closed field $\mathbb{F}$ of characteristic zero with respect to a nilpotent subalgebra $\mathfrak{h}$ is the generalized weight space decomposition with respect to the adjoint representation. That is,

$$\mathfrak{g} = \bigoplus_{\alpha \in \mathfrak{h}^*} \mathfrak{g}_{\alpha}^{\mathfrak{h}}$$

and it has the property that

$$(4.5.1) \quad [\mathfrak{g}_{\alpha}^{\mathfrak{h}}, \mathfrak{g}_{\beta}^{\mathfrak{h}}] \subseteq \mathfrak{g}_{\alpha+\beta}^{\mathfrak{h}}$$

It is important to make the distinction between the generalized weight space decomposition and the generalized root space decomposition; “we will see its convenience in later lectures, as we try to better understand the functionals α appearing in the decomposition”.

Exercise 4.6. Take $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{F})$ and $\mathfrak{h} = \{\text{diagonal matrices}\}$. Find the generalized weight space decomposition in both the tautological and the adjoint representations, and check the inclusions 4.4.1 and 4.5.1.

5. ZARISKY TOPOLOGY AND REGULAR ELEMENTS

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra of dimension d . For every element $a \in \mathfrak{g}$ the characteristic polynomial of ad_a must be of the form

$$\det_{\mathfrak{g}}(\text{ad}_a - \lambda \text{Id}) = (-\lambda)^d + c_{d-1}(-\lambda)^{d-1} + \dots + \det(\text{ad}_a).$$

[to be honest I don't see why the constant term must be the determinant of ad_a , but OK...] since $[a, a] = 0$, $\det(\text{ad}_a) = 0$, i.e., the constant term vanishes.

Exercise 5.1. Show that c_j is a homogeneous polynomial on $\mathfrak{g}$ of degree $d - j$.

Proof. The term “polynomial on $\mathfrak{g}$ ” is not clear. But we follow [Kac10] to interpret this as being a polynomial on the a^i where $a = a^i X_i$ for some basis X_i of $\mathfrak{g}$.

Then we just notice that denoting $[X_i, X_j] := L_{ij}$ we have

$$[a, X_j] = [a^i X_i, X_j] = a^i [X_i, X_j] = a^i L_{ij}$$

which is a linear combination of the a^i . Then the matrix representation of ad_a in terms of the basis X_i is

$$[\text{ad}_a] = \begin{pmatrix} a^1 L_{11}^1 & \dots & a^1 L_{1n}^1 \\ \vdots & & \vdots \\ a^1 L_{i1}^n & \dots & a^1 L_{in}^n \end{pmatrix}$$

subtracting λId and taking determinant we obtain that a term with λ^k would have for coefficient a product of k of the linear combinations $a^i L_{i\ell}^j$ for varying j and ℓ . Such a product is understood to be a homogeneous polynomial of degree k in $\mathfrak{g}$. $\square$

Definition 5.2. I think this goes for any Lie algebra $\mathfrak{g}$:

- The *rank* of $\mathfrak{g}$ is the smallest integer r such that $c_r(a)$ is not the zero polynomial on $\mathfrak{g}$.
- An element $a \in \mathfrak{g}$ is called *regular* if $c_r(a) \neq 0$.
- The *discriminant* of $\mathfrak{g}$ is the nonzero polynomial $c_r(a)$ of degree $d-r$, what?

Explanation: we compute the polynomial characteristic w.r.t. ad_a for every $a \in \mathfrak{g}$. Express this polynomial as a polynomial with coefficients in $\mathfrak{g}[t]$, (here's the question: this polynomial seems to me to be a polynomial in $\mathbb{F}$.) This polynomial does not have a constant term. But what is the next smallest-degree monomial? 1? 2? That's the rank of the Lie algebra.

6. CARTAN SUBALGEBRA

Recall that $\mathfrak{h} \subset \mathfrak{g}$ is an ideal if $[a, \mathfrak{h}] \subset \mathfrak{h}$ for all $a \in \mathfrak{g}$. Maybe $\mathfrak{h}$ is not an ideal, but we can consider the largest subalgebra of $\mathfrak{g}$ where $\mathfrak{h}$ is an ideal. This is called the normalizer.

Definition 6.1. Let $\mathfrak{h}$ be a subalgebra of a Lie algebra $\mathfrak{g}$. The *normalizer* of $\mathfrak{h}$ is $N_{\mathfrak{g}}(\mathfrak{h}) := \{a \in \mathfrak{g} \mid [a, \mathfrak{h}] \subset \mathfrak{h}\}$

I think behind the Cartan subalgebra is that, roughly, taking product with anything not in $\mathfrak{h}$ leaves $\mathfrak{h}$.

Definition 6.2. A *Cartan subalgebra* of a Lie algebra $\mathfrak{g}$ is a subalgebra $\mathfrak{h}$, satisfying the following two conditions:

- (1) $\mathfrak{h}$ is a nilpotent Lie algebra,
- (2) $N_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$.

Proposition 6.3. Let $\mathfrak{g} \subset \mathfrak{gl}_n(\mathbb{F})$ be a subalgebra containing a diagonal matrix $a = \text{diag}(a_1, \dots, a_n)$ with distinct a_i , and let $\mathfrak{h}$ be the subspace of all diagonal matrices in $\mathfrak{g}$. Then $\mathfrak{h}$ is a Cartan subalgebra (of $\mathfrak{g}$ I guess).

Idea of proof. Consider the following illustrative example:

$$\begin{aligned} & \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} - \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ &= \begin{pmatrix} a\lambda_1 & b\lambda_2 \\ c\lambda_3 & d\lambda_2 \end{pmatrix} - \begin{pmatrix} \lambda_1 a & \lambda_1 b \\ \lambda_2 c & \lambda_2 d \end{pmatrix} \\ &= \begin{pmatrix} 0 & b\lambda_2 - \lambda_1 b \\ \lambda_3 a - \lambda_2 c & 0 \end{pmatrix} \end{aligned}$$

which says that any non-diagonal matrix would escape $\mathfrak{h}$. $\square$

Theorem 6.4 (Cartan). Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over an algebraically closed field $\mathbb{F}$. Let $a \in \mathfrak{g}$ be a regular element (which exists since $\mathbb{F}$ is infinite), and let $\mathfrak{g} = \bigoplus_{\lambda \in \mathbb{F}} \mathfrak{g}_{\lambda}^a$ be the generalized eigenspace decomposition of $\mathfrak{g}$ with respect to ad_a . Then $\mathfrak{g}_0^a$ is a Cartan subalgebra.

Idea of proof. First recall what is $\mathfrak{h} = \mathfrak{g}_0^a$, the set of elements $b \in \mathfrak{g}$ such that $\text{ad}_a^N(b) = 0$. $\square$

Remark 6.5. The rank of a Lie algebra is the dimension of $\mathfrak{g}_0^a = \mathfrak{h}$.

Proposition 6.6. *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over an algebraically closed field $\mathbb{F}$ of characteristic zero and let $\mathfrak{h} \subset \mathfrak{g}$ be a Cartan subalgebra. Then $\mathfrak{g}_0 = \mathfrak{h}$ in the generalized weight space decomposition.*

Idea of proof. Engel's theorem. . . □

7. SEMISIMPLE LIE ALGEBRAS

Recall that the root space decomposition works best for semisimple finite-dimensional Lie algebras!

Exercise 7.1. Let $\mathfrak{g}$ denote a Lie algebra over a field of characteristic 0. Show that $\mathfrak{g}$ is semisimple if and only if the Killing form on $\mathfrak{g}$ is nondegenerate.

Definition 7.2. A *radical* $R(\mathfrak{g})$ of a finite-dimensional Lie algebra $\mathfrak{g}$ is a solvable ideal of $\mathfrak{g}$ of maximal possible dimension.

Proposition 7.3. *The radical ideal of $\mathfrak{g}$ contains any solvable ideal of $\mathfrak{g}$ and is unique.*

If $\mathfrak{g}$ is a finite dimensional solvable Lie algebra, then $R(\mathfrak{g}) = \mathfrak{g}$. The opposite case is when $R(\mathfrak{g}) = 0$.

Definition 7.4. A finite-dimensional Lie algebra $\mathfrak{g}$ is called *semisimple* if $R(\mathfrak{g}) = 0$.

In [Kac10, Lecture 11] we have the tools needed for a characterization of semisimple Lie algebras in terms of the Killing form:

Theorem 7.5 (Cartan). *Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field of characteristic 0. Then the Killing form on $\mathfrak{g}$ is non-degenerate if and only if $\mathfrak{g}$ is semisimple. Moreover, if $\mathfrak{g}$ is semisimple and $\mathfrak{a} \subset \mathfrak{g}$ is an ideal, then the restriction of the Killing form to $\mathfrak{a}$, $K|_{\mathfrak{a} \times \mathfrak{a}}$, is also nondegenerate and coincides with the Killing form of $\mathfrak{a}$.*

I think we need semisimplicity, i.e. nondegeneracy of the Killing form for the following definition:

Definition 7.6. Given a set of simple roots $\Pi = \{\alpha_1, \dots, \alpha_\ell\}$, we define the *coroot*

$$(7.6.1) \quad \alpha_i^\vee = \frac{2}{(\alpha_i, \alpha_i)} \nu^{-1}(\alpha_i)$$

where $\nu : \mathfrak{h} \rightarrow \mathfrak{h}^*$ is the pairing induced by the Killing form.

8. $\mathfrak{g}$ -MODULES

Definition 8.1. Let $\mathfrak{g}$ be a Lie algebra. A $\mathfrak{g}$ -*module* (also called a *representation* of $\mathfrak{g}$) is a vector space V and a homomorphism

$$\rho : \mathfrak{g} \rightarrow \text{End}(V)$$

of Lie algebras, i.e.,

$$\rho([x, y]) = \rho(x)\rho(y) - \rho(y)\rho(x).$$

A vector space $W \subset V$ such that $vW \subset W$ for all $x \in \mathfrak{g}$ is called a $\mathfrak{g}$ -*submodule*.

Definition 8.2. An *irreducible $\mathfrak{g}$ -module* is one that has no $\mathfrak{g}$ -submodules other than the trivial ones (namely, 0 and V itself).

It is possible to classify and describe the irreducible finite-dimensional $\mathfrak{g}$ -modules. This is in contrast with all the modules of a Lie algebra $\mathfrak{g}$, which is “impossible in general”.

9. VERMA MODULE

Let M be a $\mathfrak{g}$ -module. Recall Definition 3.1: let $\mathfrak{h}$ be a Lie algebra, $\pi : \mathfrak{h} \rightarrow \mathfrak{gl}_V$ a representation of $\mathfrak{h}$ and $\lambda \in \mathfrak{h}^*$. The *weight space* of $\mathfrak{h}$ attached to λ is

$$V_\lambda^\mathfrak{h} := \{v \in V \mid \pi(h)v = \lambda(h)v, \forall h \in \mathfrak{h}\}.$$

We have that a weight space in M is a nonzero subspace $M_\lambda \subset M$ such that

$$hv = \lambda(h)v \quad \forall h \in \mathfrak{h} \subset \mathfrak{g} \text{ and } v \in M,$$

and in this case we call λ a weight. If M is a direct sum of weight spaces, we call it a *weight module*.

The *support* of a weight module M is

$$(9.0.1) \quad \text{supp}(M) := \{\lambda \in \mathfrak{h}^* : M_\lambda \neq 0\}$$

We call elements of M_λ *vectors of weight λ* .

Definition 9.1. $\Lambda \in \mathfrak{h}^*$ is a *highest weight* of M if $\Lambda \in \text{supp}(M)$ and for all positive roots $\alpha \in \Delta_+$ we have that $\Lambda + \alpha \notin \text{supp}(M)$.

Exercise 9.2. The highest weight of an irreducible $\mathfrak{g}$ -module is unique.

Proof.

□

Definition 9.3. Let $\Lambda \in \mathfrak{h}^*$ (a highest weight, I suppose). The *Verma module* $M(\Lambda)$ of highest weight Λ is

$$M(\Lambda) = U(\mathfrak{g}) \otimes_{U(\mathfrak{h} \oplus \mathfrak{n}_+)} \mathbb{C}_\Lambda,$$

where $\mathbb{C}_\Lambda = \mathbb{C}v_\Lambda$ is the 1-dimensional $(\mathfrak{h} \oplus \mathfrak{n}_+)$ -module defined by $\mathfrak{n}_+v_\Lambda = 0$ and $hv_\Lambda = \langle \Lambda, h \rangle v_\Lambda$ for all $h \in \mathfrak{h}$.

Long story short, by the PBW Theorem we have an isomorphism $U(\mathfrak{n}_-) \rightarrow M(\Lambda)$ of $U(\mathfrak{n}_-)$ modules.

10. WEYL CHARACTER FORMULA

There are at least two Weyl character formulas. This one is [?, Theorem 10.14]:

Theorem 10.1 (Weyl Character Formula). *If (π, V_μ) is an irreducible representation of $\mathfrak{g}$ with highest weight μ , then*

$$(10.1.1) \quad \chi_\pi(H) = \frac{\sum_{w \in W} \det(w) e^{\langle w \cdot (\mu + \delta), H \rangle}}{\sum_{w \in W} \det(w) e^{\langle w \cdot \delta, H \rangle}}$$

for all $H \in \mathfrak{h}$ for which the denominator is nonzero.

And this one is [Kac10, Theorem 25.3]:

Theorem 10.2 (Weyl Character Formula). *Let $R = \prod_{\alpha \in \Delta_+} (1 - e^{-\alpha})$. If $\Lambda \in P_+$, then*

$$(10.2.1) \quad e^\rho \text{Rch}L(\Lambda) = \sum_{w \in W} \det w e^{w(\Lambda + \rho)}$$

Exercise 10.3. Let $\mathfrak{g} = \mathfrak{sl}_3$. Compute $\text{ch}_{L(\lambda)}$ for some λ ,

- (1) $\lambda = 0$,
- (2) $\lambda = \omega_1$,
- (3) $\lambda = \omega_1 + \omega_2$,

etc.

Proof. No matter what formula I use, I must compute the determinants of the Weyl reflections and the ρ (which I think is δ for Hall), which is just the sum of all the fundamental weights, which in turn are the dual of the coroot vectors. Finally applying the reflections to ρ and to $\lambda + \rho$ would yield the result.

- (1) (Find roots of $\mathfrak{sl}_3$.) We look for $\alpha \in \mathfrak{h}^* = \{\text{diagonal matrices}\}$ such that

$$[H, E_{ij}] = \alpha(H)E_{ij} \quad \forall H \in \mathfrak{h}$$

where E_{ij} is the matrix that has zero in every entry but in the (i, j) -th where it has a 1. One obtains that $[H, E_{ij}] = (h_i - h_j)E_{ij}$, so that the roots are $\alpha_{ij}(h) = h_i - h_j$.

- (2) (Compute the Killing form.) Recall that by definition $\kappa(H, H') = \text{Tr}(\text{ad}_H \text{ad}_{H'})$. But in this case, we have $\kappa(H, H') = \text{Tr}(HH')$ because $\text{ad}_H \text{ad}_{H'} = [H, [H', A]]$

The first thing Jethro did was to consider a basis

$$H_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad H_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

and then compute the Killing form via

$$H_1 H_1 = 2, \quad H_1 H_2 = -1, \quad H_2 H_1 = -1, \quad H_2 H_2 = 2.$$

- (3) (Find the dual basis vectors.) Now we use the musical isomorphism $\nu : \mathfrak{h} \rightarrow \mathfrak{h}^*$, $H \mapsto (-, H)$ to find

$$\nu(H_1) =$$

- (4) (Find fundamental weights.) It looks like $\alpha_1 = 2\omega_1 - \omega_2$ and $\alpha_2 = 2\omega_2 - \omega_1$. This says $\omega_1 = 2\omega_2 - \alpha_2 = 2(2\omega_1 - \alpha_1) - \alpha_2$ so $3\omega_1 = 2\alpha_1 + \alpha_2$. And then $\omega_2 = 2(\frac{2}{3}\alpha_1 + \frac{1}{3}\alpha_2) - \alpha_1 = \frac{1}{3}(\alpha_1 + 2\alpha_2)$.
- (5) (Find ρ .) Then $\rho = \omega_1 + \omega_2 = \alpha_1 + \alpha_2$.
- (6) (Find Weyl reflections.) To find Weyl reflections I need to compute first the numbers (α_i, α_j) . I think this is still not very clear — have to use Killing form? But OK, I know we must have $(\alpha_i, \alpha_i) = 2$, and the both of the crossed ones give -1 . This should give by bilinearity the values on ω_i also.
- (7) (Find determinants of Weyl reflections.) I guess for this I have to evaluate the reflections in each of the w
- (8) (Evaluate ρ and $\lambda + \rho$ on Weyl reflections.)
- (9) (Find the Weyl character.)

What's funny in this whole story is that in the end we can just find the Weyl character geometrically. For every $\lambda = m\omega_1 + n\omega_2$ I want to compute its orbit under the Weyl group W . This means

I would start by finding the determinant of the Weyl reflections... □

REFERENCES

- [Kac10] Victor Kac, *Lecture notes of 18.745 – introduction to lie algebras (fall 2010)*, <https://math.mit.edu/classes/18.745/classnotes.html>, 2010, Lecture notes, MIT.